Shayantan Biswas

B.Tech - Computer Science and Engineering
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Educational Qualification

Graduation

SEM	1	2	3	4	5	6	7	8
SGPA	9.04	8.45	8.31	8.55	8.78	9.00	[SGPA7]	[SGPA8]
CGPA	9.04	8.77	8.59	8.58	8.62	8.69	[CGPA7]	[CGPA8]

Higher Secondary & Secondary

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2022	Class XII	WBCHSE	93
2020	Class X	WBBSE	90

Achievements

- Selected for **Amazon ML Summer School 2026** through Unstop, gaining the opportunity to learn advanced Machine Learning concepts from Amazon industry experts.
- Practices Data Structures and Algorithms daily through LeetCode to strengthen problem-solving and algorithmic thinking skills.
- Secured an All India Rank of 9614 in JEE Mains.
- Participated in ISRO Hackathon, Walmart Hackathon, and Smart India Hackathon (SIH) 2024, contributing to innovative problem-solving and rapid prototyping in team-based environments.

Skills Summary

- **Programming Languages:** C/C++, Python, JavaScript, HTML, CSS
- **Frameworks:** React.js , Express.js , Flask
- **Libraries:**Python Libraries (NumPy, Pandas, Matplotlib)
- **Web Development Tools:** Node.js, VS Code, Git, GitHub
- **Database:** MongoDB, MySQL
- **AI & Machine Learning:** Machine Learning, Deep Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG)
- **Core CS Concepts:** Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems , Software Engineering, Computer Network

Projects

Full Stack Food Application — React.js, Node.js, Express.js, TailwindCSS, MongoDB

- Designed and developed a full-stack food ordering web application featuring secure user authentication, interactive menu browsing, dynamic cart functionality, and order management. Integrated a Flask-based backend to incorporate AI-driven features, enhancing personalization and system intelligence. Built using modern JavaScript frameworks, the platform delivers a seamless, responsive user experience with real-time updates. Ensured efficient front-end and back-end integration along with an intuitive, user-friendly interface for end-to-end food ordering.
- **Repository Link:** **Full Stack Food Application Repository**

GATEMIND-AI — React.js, Flask, MongoDB, LangChain, RAG, Docker, AWS

- Developed an AI-powered GATE preparation platform featuring JWT authentication, OTP verification, mock tests, and subject-wise performance analytics. Implemented a personalized RAG chatbot with hybrid semantic–lexical retrieval, document embeddings, conversation history, and citation-based responses. Designed a modular Flask backend using SOLID principles and multiple LLD design patterns, and deployed the React frontend on Vercel and Dockerized backend on AWS EC2 using GitHub Actions CI/CD.

- **Repository Link: GATEMIND-AI Repository**

Explainable AI for Plant Seedling Classification — PyTorch, LIME, SHAP, Scikit-learn, Matplotlib, Pandas

- Conducted a research project on plant seedling image classification using deep learning and Explainable AI (XAI) techniques. Applied interpretation methods to analyze model predictions and assess feature importance and performance.
- **Repository Link: Plant Seedling XAI Repository**

Experience

CMATER Lab ,Jadavpur University Dec 2024 – Jan 2025
Winter Research Intern – Deep Learning for PV Cell Generation Forecasting Remote

- Worked under Prof. Debotosh Bhattacharjee on PV power forecasting using deep learning; applied feature selection and ensemble models on time-series data, achieving 98% accuracy.
- Completion Certificate: View Certificate

Indian Institute of Technology , Guwahati May 2025 – July 2025
Summer Research Intern – Tabular Deep Learning, Self-Supervised Learning Remote

- Worked under Prof. Arijit Sur on a project exploring and evaluating self-supervised learning models on structured datasets, achieving up to 70% accuracy.
- Completion Certificate: View Certificate

Indian Institute of Technology , Hyderabad May 2026 – Jul 2026
Summer Research Intern (SRFP) – Clustering , Multithreading On-Site

- Worked under Prof. Sathya Peri on **incremental clustering algorithms**, implementing and optimizing solutions in **C++**.
- Applied **distance-based outlier detection** to remove noisy radar signal points before clustering, improving the reliability of emitter identification.

Position of Responsibility

IEEE Computer Science Society, NIT Durgapur — Head of Data Science & Treasurer Feb 2025 – Present

- Led and organized flagship technical events like Hack Hustle (24hr hackathon), ENVISION (autonomous bot challenge), PCB Decoded, and
- Introduced and organized Epoch, a 5-day AI/ML workshop and hackathon for first-year students, sponsored by GeeksforGeeks (GFG).